

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 8	Team No- 12	Project Title- Smart Restaurant Ordering & Kitchen Display System
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1. Abstract

Traditional dine-in restaurants often face delays and errors caused by manual order-taking, illegible handwritten tickets, and miscommunication between the waitstaff and the kitchen. This project proposes a Smart Restaurant Ordering & Kitchen Display System that digitizes the entire order lifecycle, from table-side order placement to real-time kitchen preparation tracking. Customers or waitstaff place orders through a simple digital interface, which instantly transmits the order details to a Kitchen Display System (KDS) replacing paper tickets with organized, color-coded digital order cards. Orders are automatically queued, timestamped, and updated as they move through stages such as received, in-preparation, and ready-to-serve, giving kitchen staff a clear, prioritized view of pending work. The system also maintains a live status dashboard for the front-of-house staff, reducing the need for verbal follow-ups and cutting down order-serving time. A backend database stores menu items, order history, and preparation times, enabling analytics on peak hours, popular dishes, and kitchen efficiency. By eliminating manual paperwork, reducing order errors, and improving coordination between service and kitchen staff, the proposed system enhances overall restaurant efficiency, customer satisfaction, and staff productivity, while providing a scalable foundation for future features such as inventory tracking and customer feedback integration.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030234	T. Karthik	
2520030239	CH.V.S. Manikanta Varma Raju	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend: React.js (customer/waiter ordering interface) and React Native (KDS tablet display)

Backend: Node.js with Express.js for REST APIs and order management logic

Database: MySQL for structured data (menu, orders, users) with Redis for order-queue caching

Real-time Communication: Socket.IO / WebSockets for instant order updates between ordering app and kitchen display

Hosting/Deployment: Docker containers deployed on AWS/Firebase Hosting

Other Tools: Postman (API testing), Git/GitHub (version control), Figma (UI/UX design)

Guide Name and Signature (remarks if any) _____